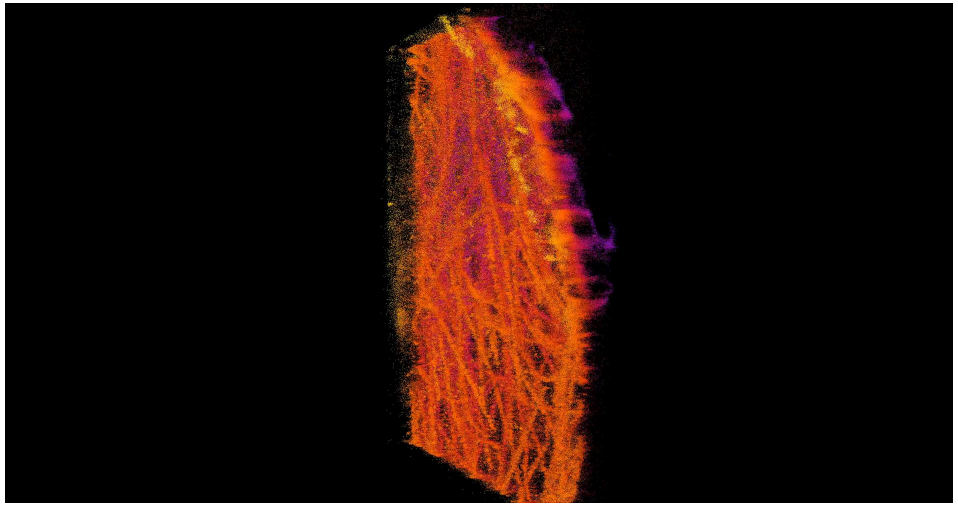
Learning-methods for single molecule localization microscopy

Romain Séailles



PDF export — this is a static version of an interactive presentation; some slides may appear incomplete or degraded.

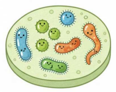
Light microscopy

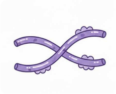
Light microscopy is a technique that uses **visible light** and a system of lenses to generate magnified images of small objects.


1 mm


100µm


10µm


1µm


100nm


Optical microscope

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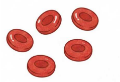
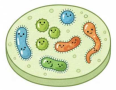
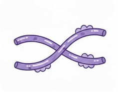

10μm


1μm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Abbe diffraction limit (1873)

The minimum resolvable distance is

$$d = \frac{\lambda}{2n\sin\theta} = \frac{\lambda}{2\,\mathrm{NA}}.$$

For a conventional setup, d $\simeq$ 200nm.

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		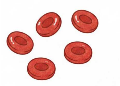	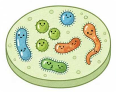	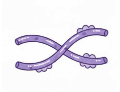
1 mm	100µm	10µm	1µm	100nm

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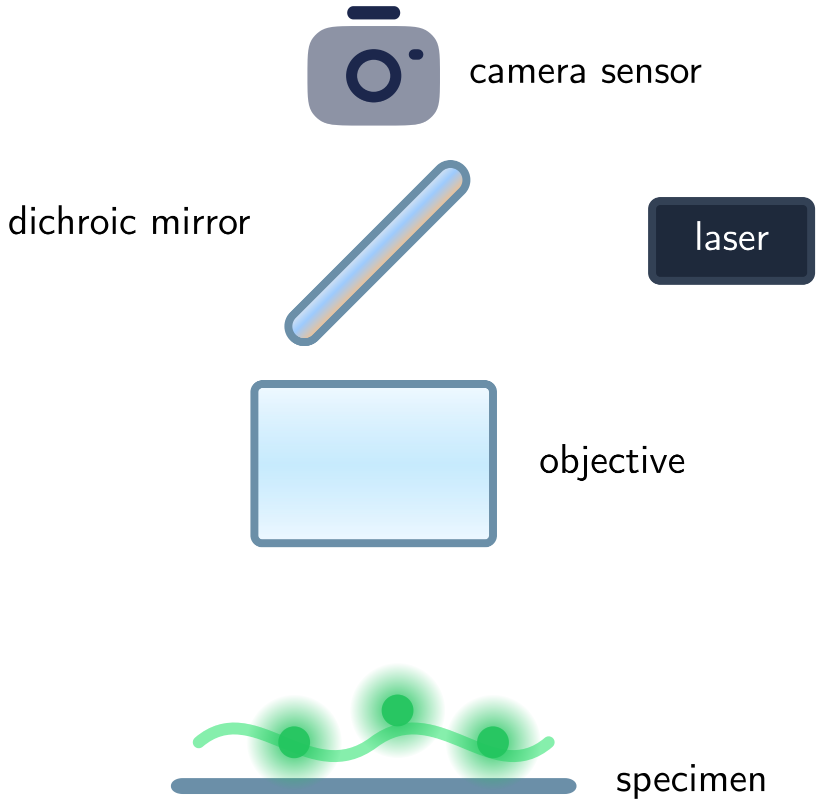
Can we circumvent this limit ?

Optical microscope

Fluorescence microscopy

Fluorescence microscopy exploits fluorophores stained on the specimen to super-resolve acquisitions.



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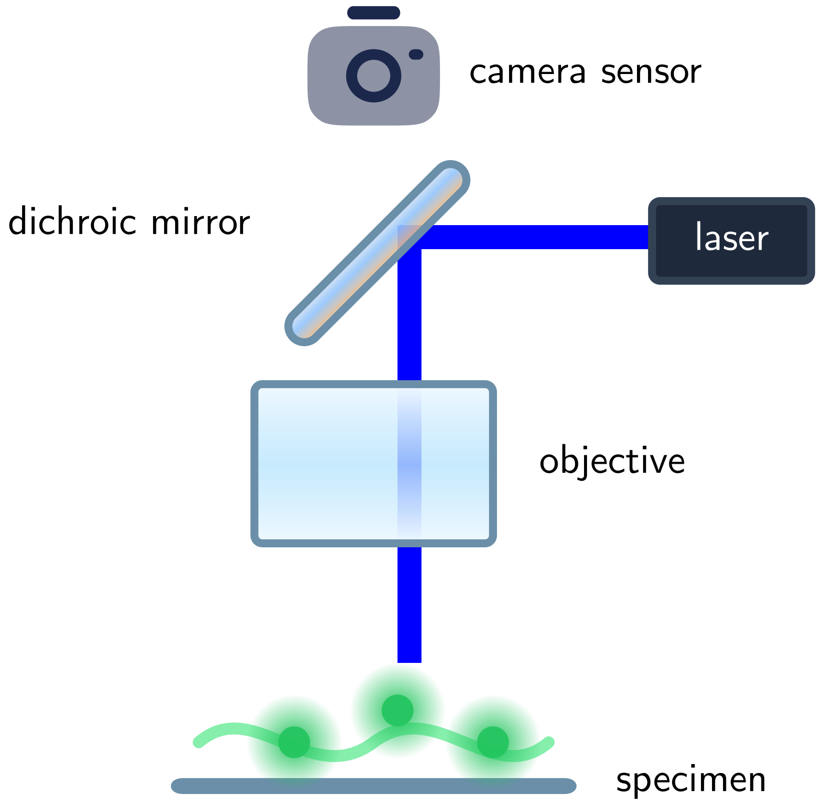

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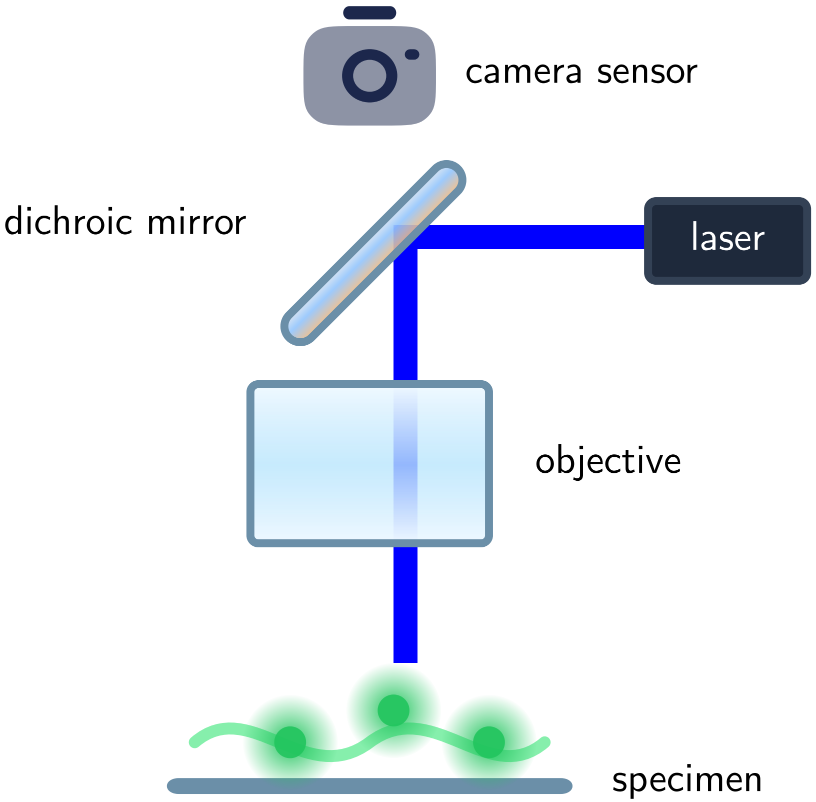
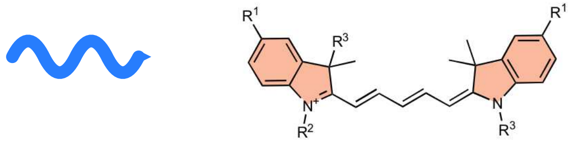


Fluorophores are fluorescent chemical compounds that once excited stochastically re-emit light at a longer wavelength.



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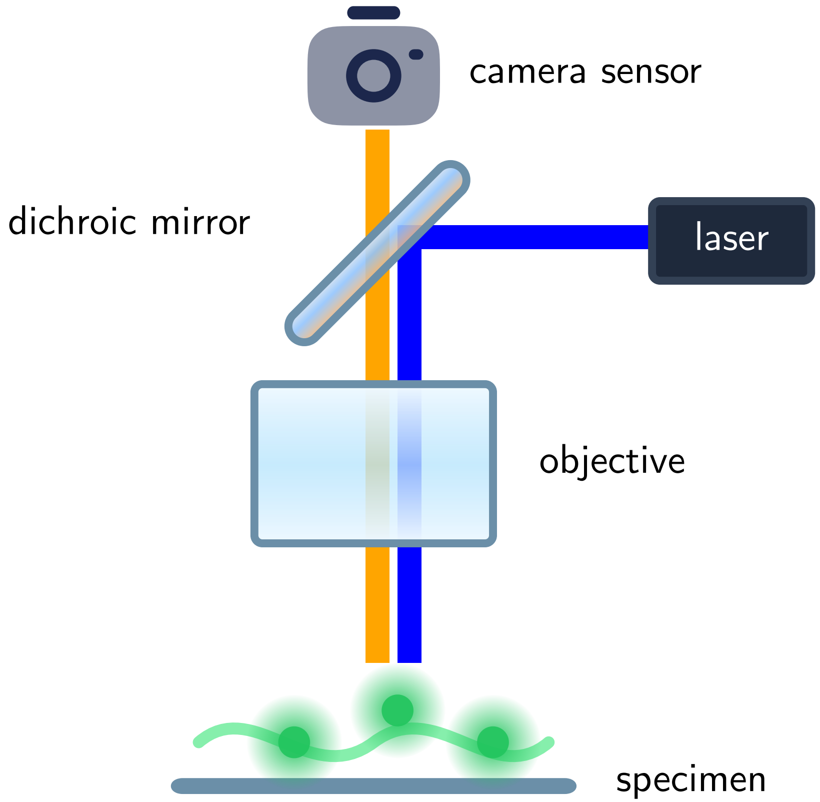

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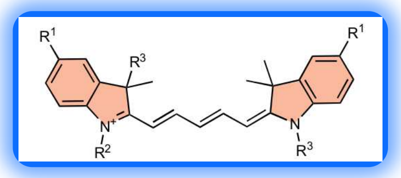

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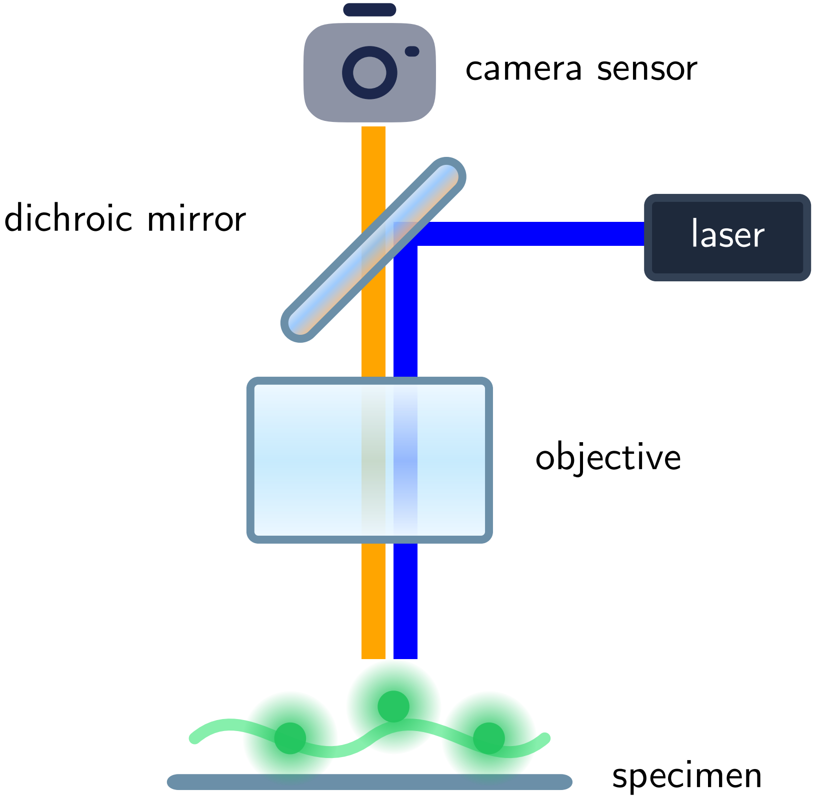

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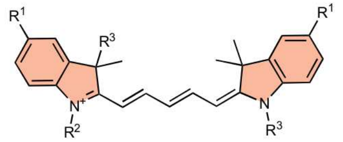
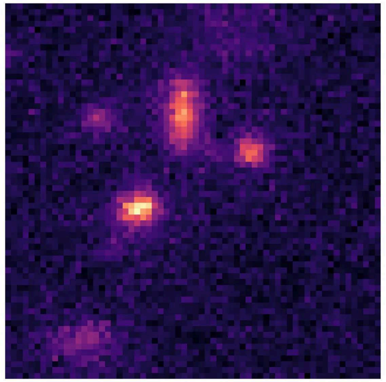


We observe a **sequence of images**, showing stochastically blinking "blobs".



3D Point spread function

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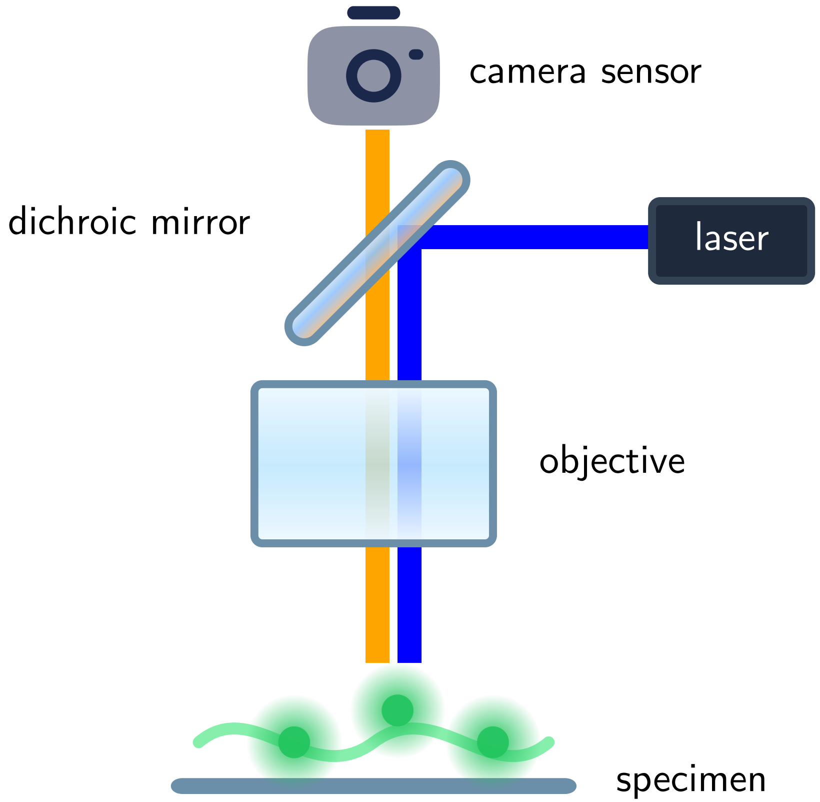
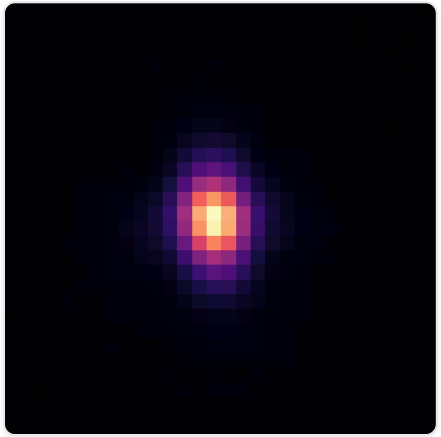


Each "blob" is the **point spread function (PSF)**: the image of a single point source formed by the microscope.



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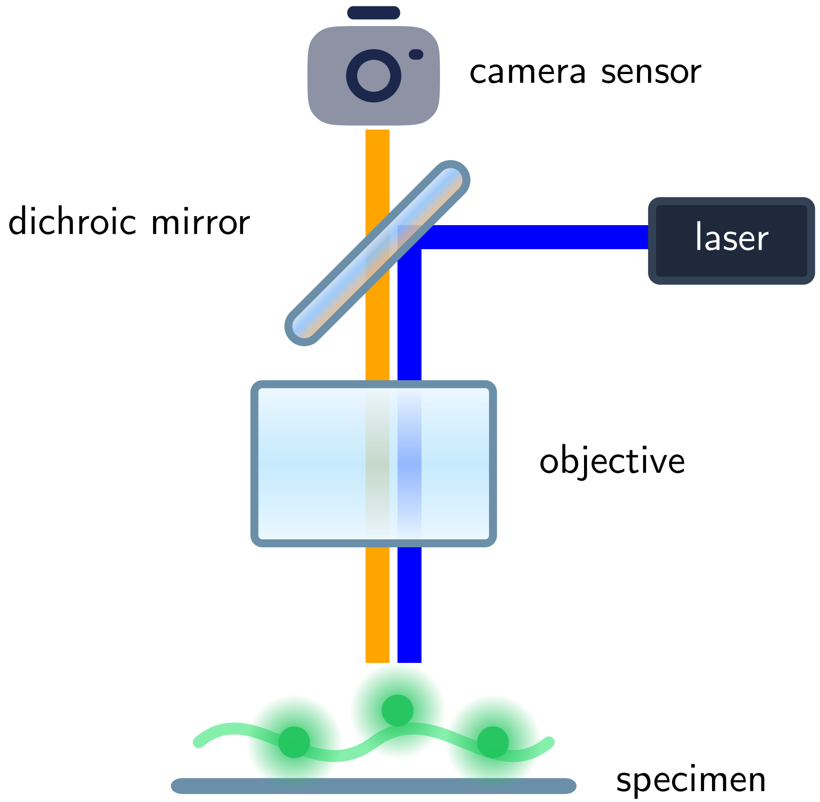

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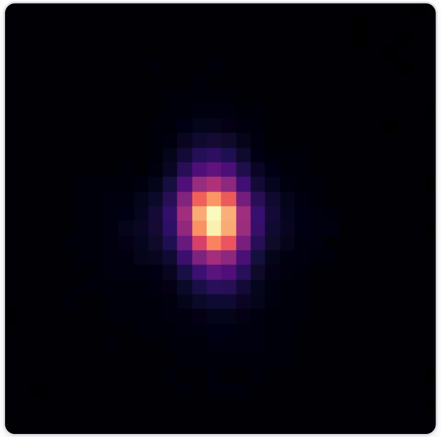


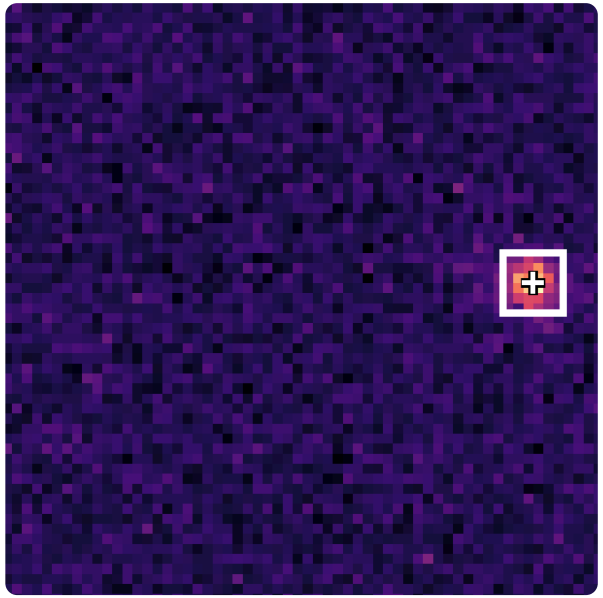
We can use **cylindrical lenses** in the objective to introduce **astigmatism**, which breaks the axial symmetry.

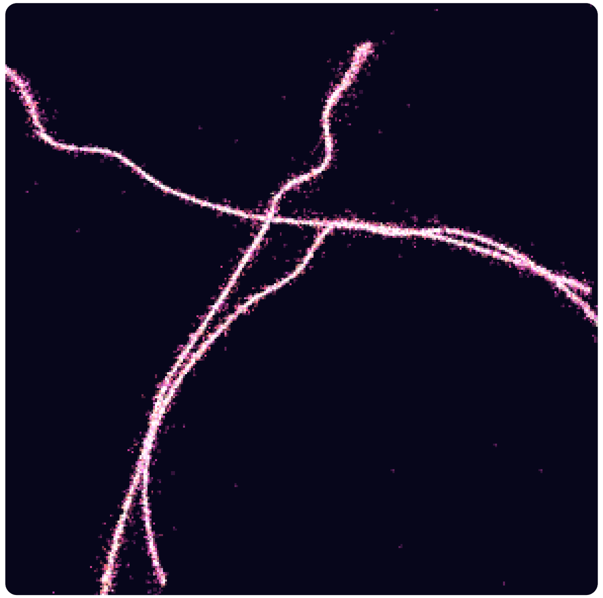
At the focal plane

Single molecule localization microscopy

Single molecule localization microscopy (SMLM) **detects** and **localizes** individual fluorophores to reconstruct a high-resolution image.

Frames $\leq$ 1





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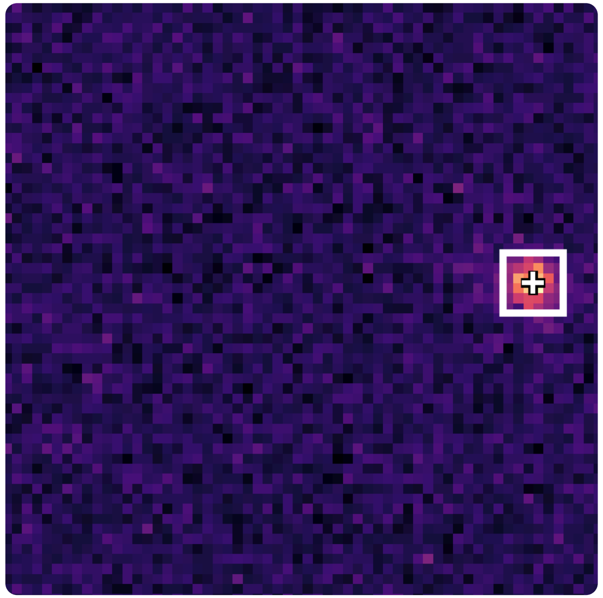

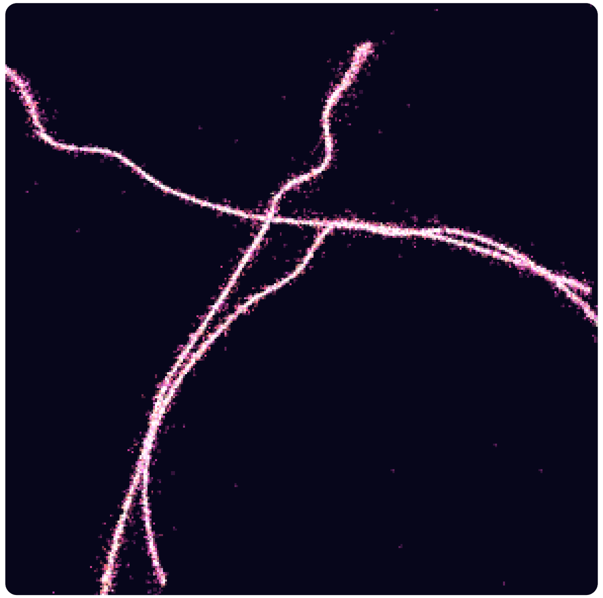

Breaks diffraction limit

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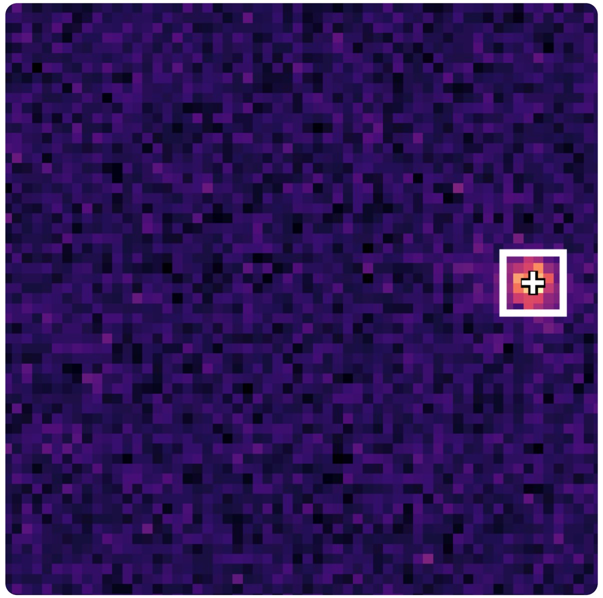

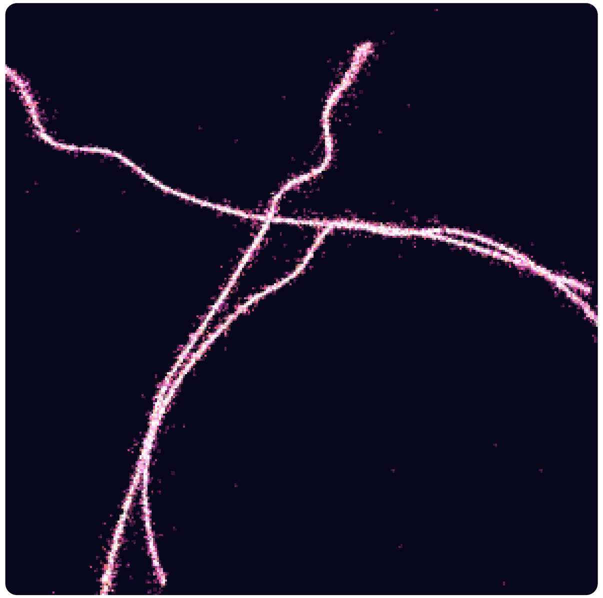

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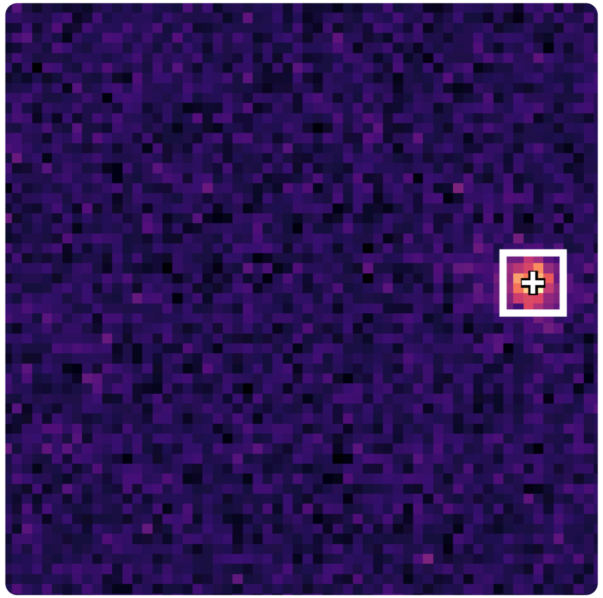

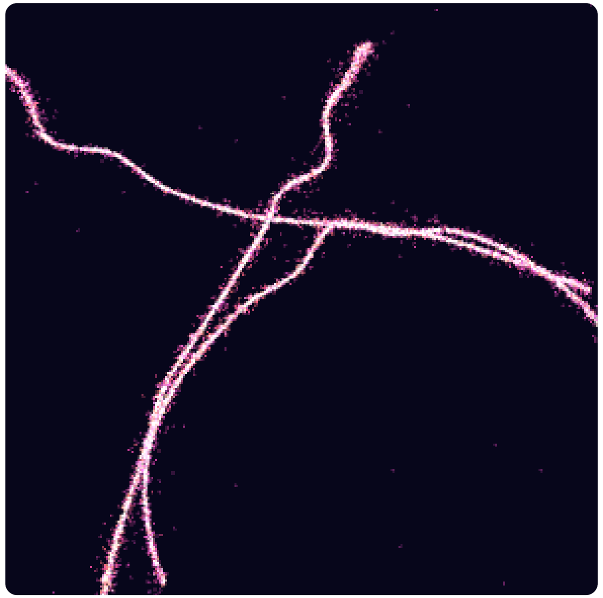

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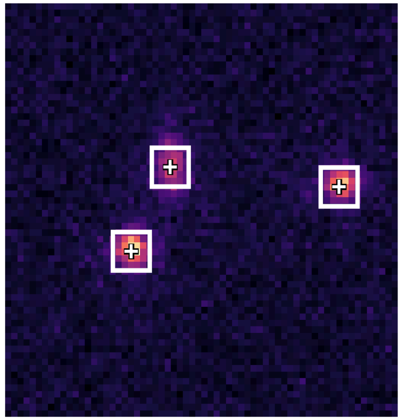
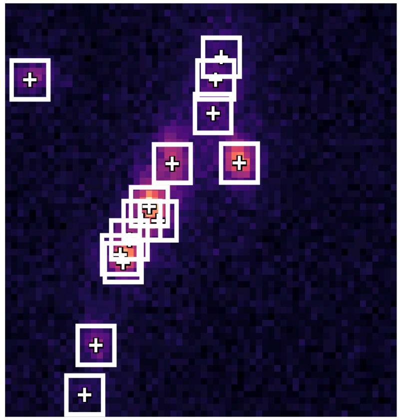
Slow acquisition 🐌

SMLM Density

Low Density: Laser power is tuned to limit the number of active fluorophores, such that **no two activations occurred simultaneously in the same diffraction-limited area** with high probability.



0.2 act/μm^2/frame



2.0 act/μm^2/frame

High Density: Overlapping activations create uncertainties in the number of fluorophores, leading to **localization errors** and a degraded reconstruction.

Single molecule localisation microscopy

Advantages

- **Excellent resolution**, $d \simeq 30$nm in a classic setup.

  Can reach 10nm if the density is low enough.
- **Less phototoxicity** for photosensitive specimens.

  (compared to active methods like STED)

Inconvenients

- Limited to "low" activation densities

  Otherwise, precise individual localization becomes impossible.
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  Imaging process can take minutes to hours.
- Not suited for the observation of fast biological processes.

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Can learning-based methods enable faster SMLM reconstruction ?

Optimal transport unlocks end-to-end learning for single-molecule localization

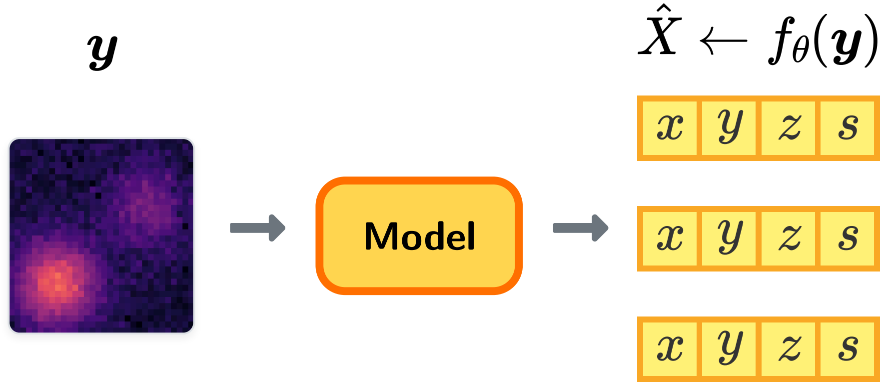
Romain Séailles, Jean-Baptiste Masson, Jean Ponce, and Julien Mairal.

2026 to appear in ICLR

Joint detection and localization framework

Given a frame $\boldsymbol{y}$, a model f_θ predicts a set of candidates $\hat{X} = \{(x_i, y_i, z_i, s_i)\}_{1 \le i \le d}$ where:

- (x_i, y_i, z_i): 3D spatial coordinates
- $s_i \in [0, 1]$: Detection confidence score



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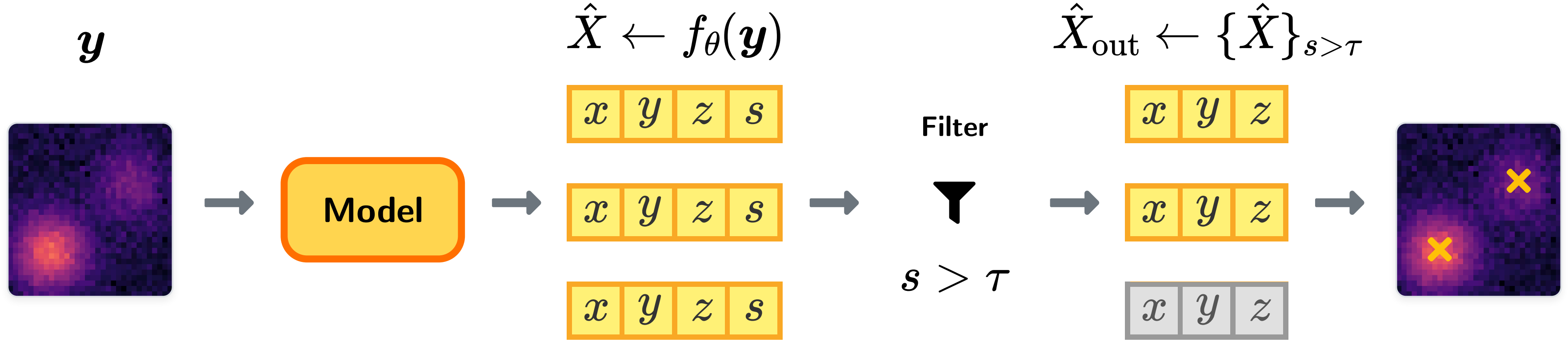

At inference, candidates are filtered by score ($s > \tau$)
$\rightarrow$ this threshold controls the Precision-Recall trade-off.

Supervised learning

Following Speiser et al. (2021), we propose to train our model using **supervised learning**.

Mathematically, we **minimize the empirical risk** via **stochastic gradient descent**:

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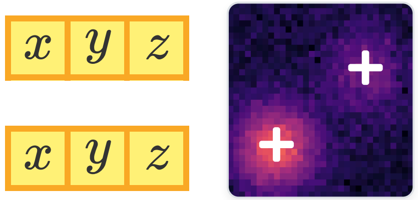

A fluorophores-image pair dataset

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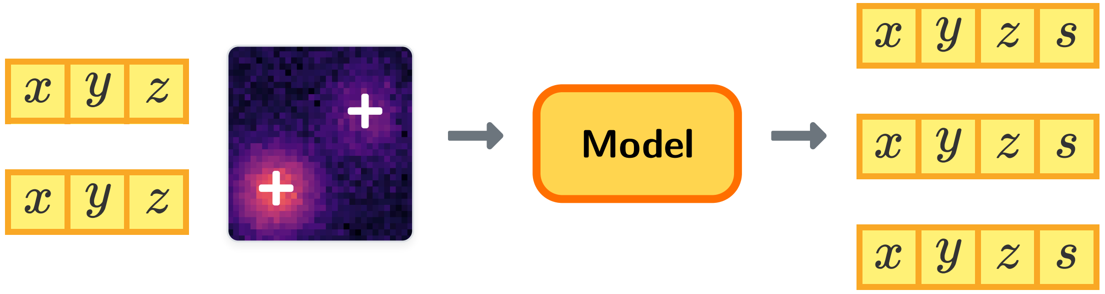

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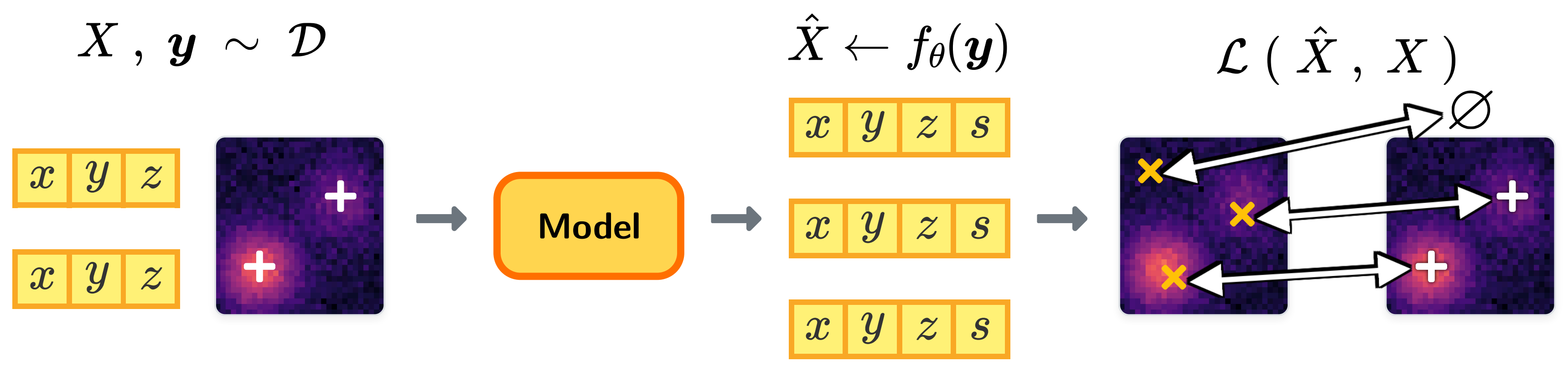

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A loss function

Synthetic data

Ground truth unavailable $\Rightarrow$ simulator built on the model of Sage et al. (2019)

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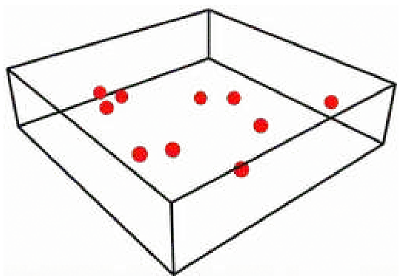

Uniform 3D spatial distribution.

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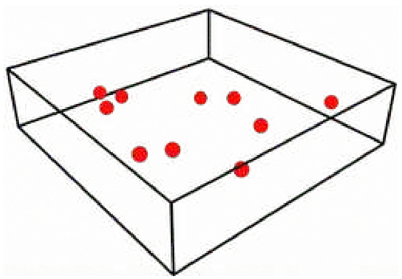

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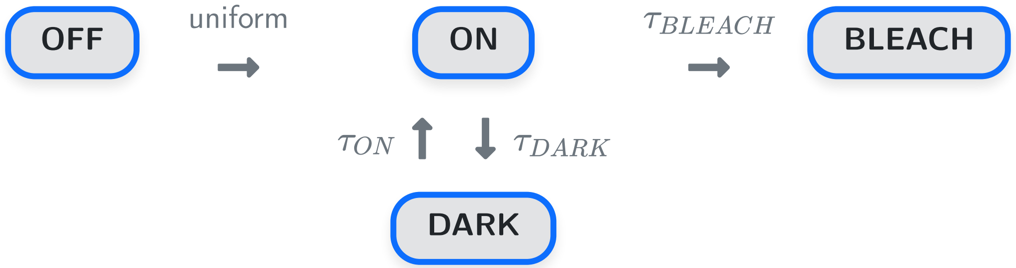

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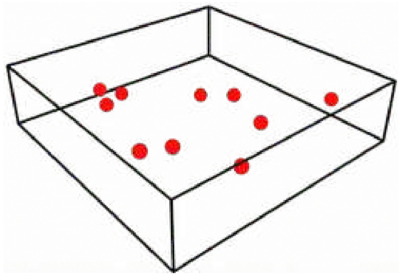

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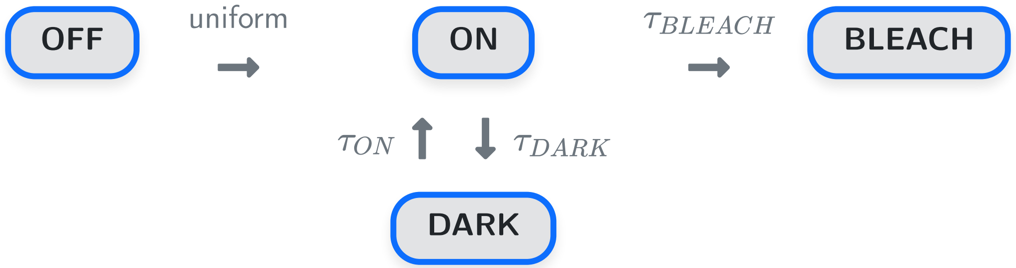

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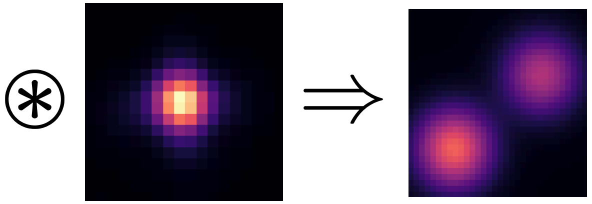

Empirical PSF calibrated on real data.

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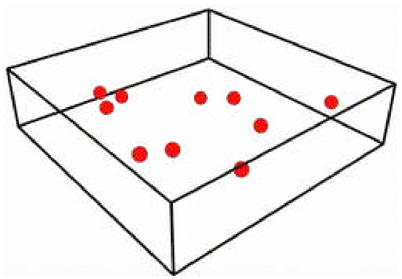

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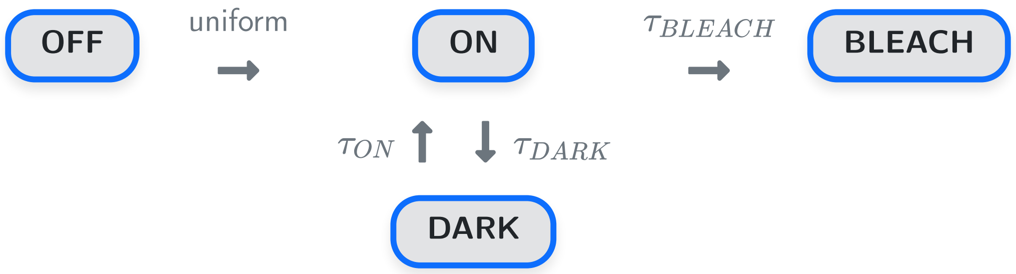

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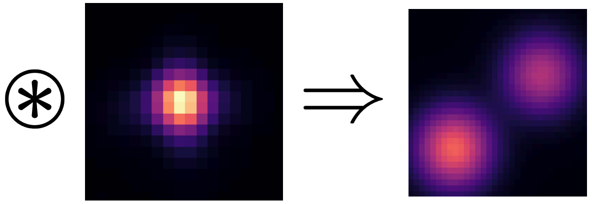

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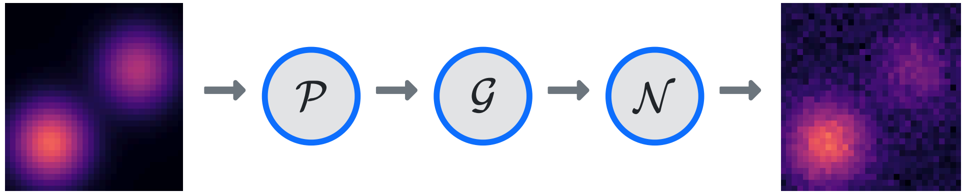
4. Camera model



Composite noise includes shot noise (Poisson dist.), amplification noise (Gamma dist.) and readout noise (normal dist.).

Model architecture

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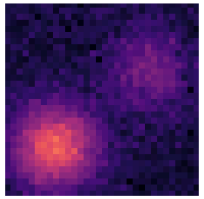

> Input image $\boldsymbol{y} \in \mathbb{R}^{H \times W}$.

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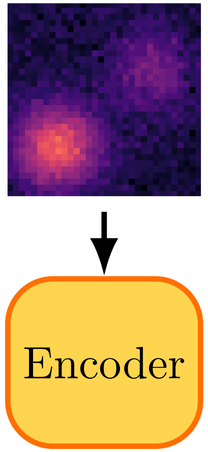

The encoder is implemented as a 2-layer U-Net, mapping $\boldsymbol{y}$ to a latent representation $\boldsymbol{z} \in \mathbb{R}^{C \times H \times W}$.

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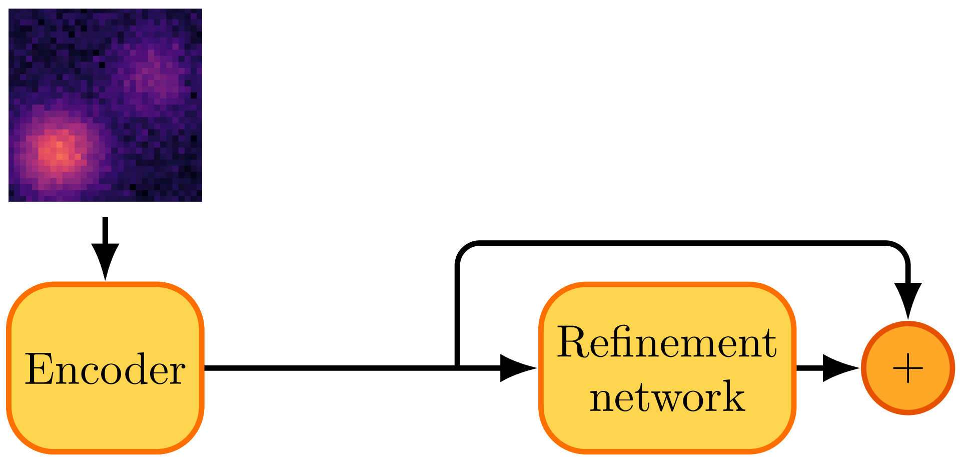

> The core network is also implemented as a 2-layer U-Net.

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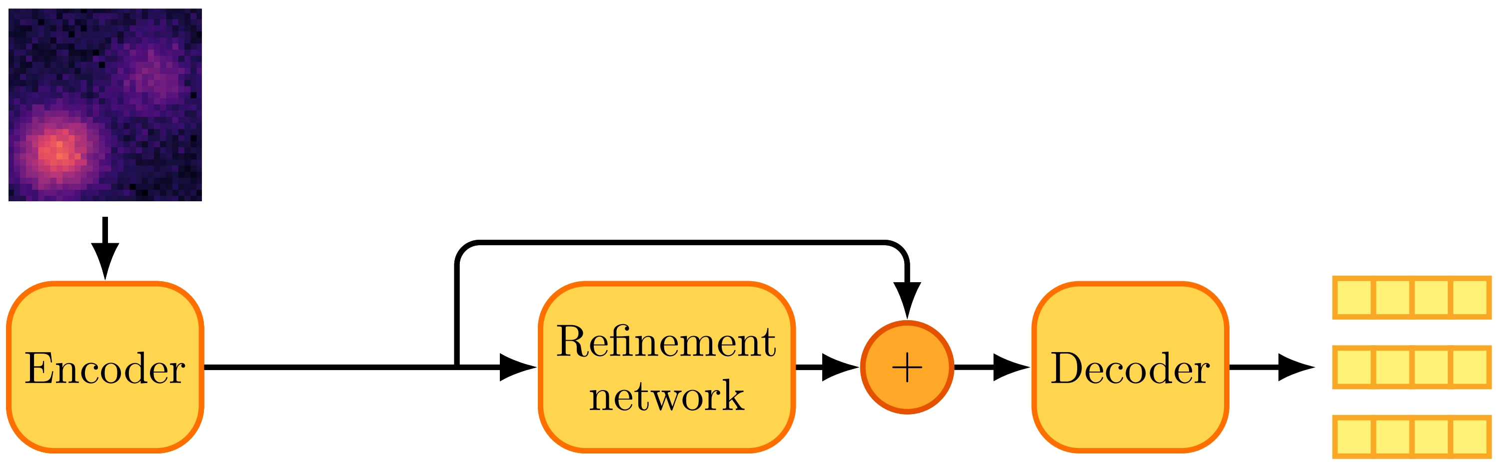

The decoder is implemented as a shallow CNN with max-pooling followed by flattening, yielding $\hat{X} \in \mathbb{R}^{d\times 4}$.

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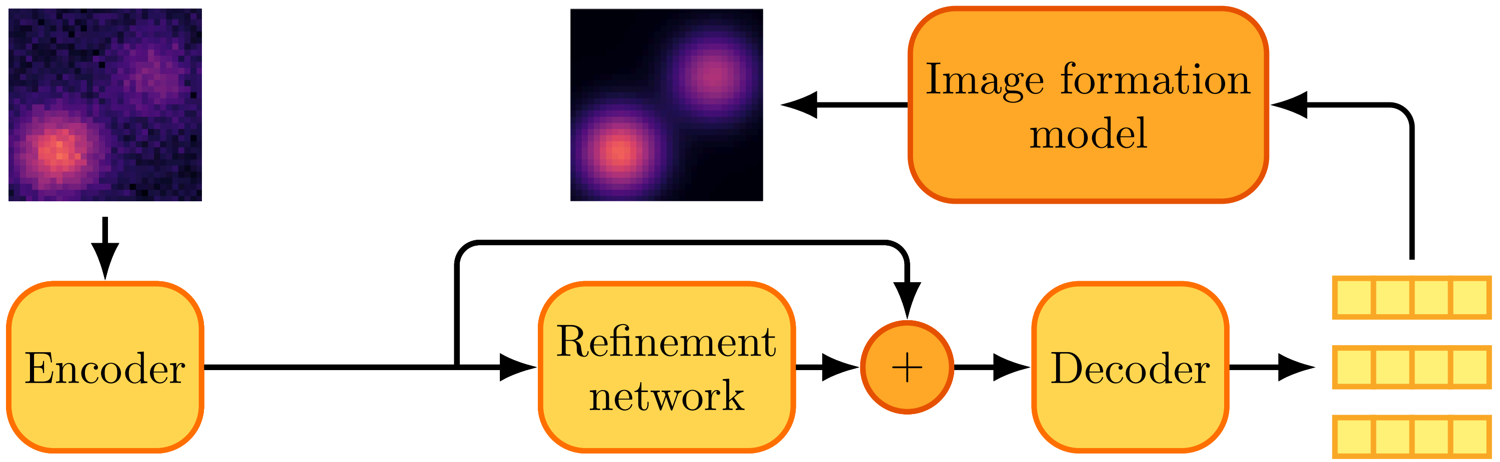

We render the output $\hat{X}$ into a new image $\hat{\boldsymbol{y}}$ with the image formation process described earlier (using relaxation tricks on detection scores to preserve differentiability).

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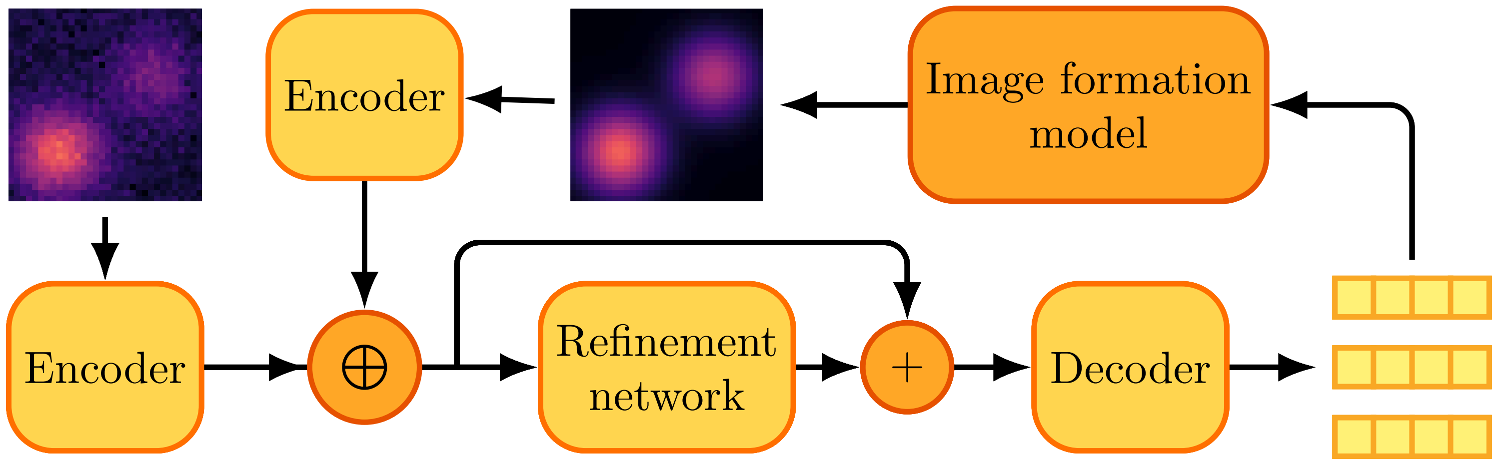

The same encoder maps the rendered image $\hat{\boldsymbol{y}}$ to a latent representation $\hat{\boldsymbol{z}} \in \mathbb{R}^{C\times H\times W}$. The original and new latents, $\boldsymbol{z}$ and $\hat{\boldsymbol{z}}$, are concatenated for refinement in the next iteration.

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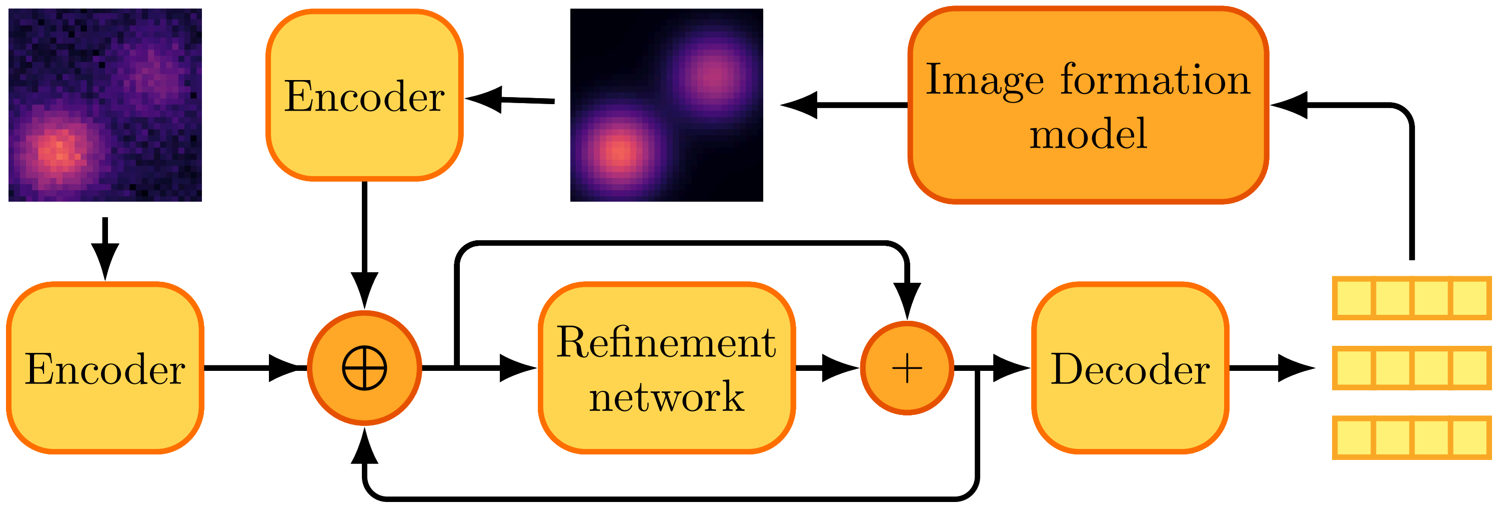

> The refinement pass, alongside the original and rendered latent representation, also needs the current latent state to effectively refine it.

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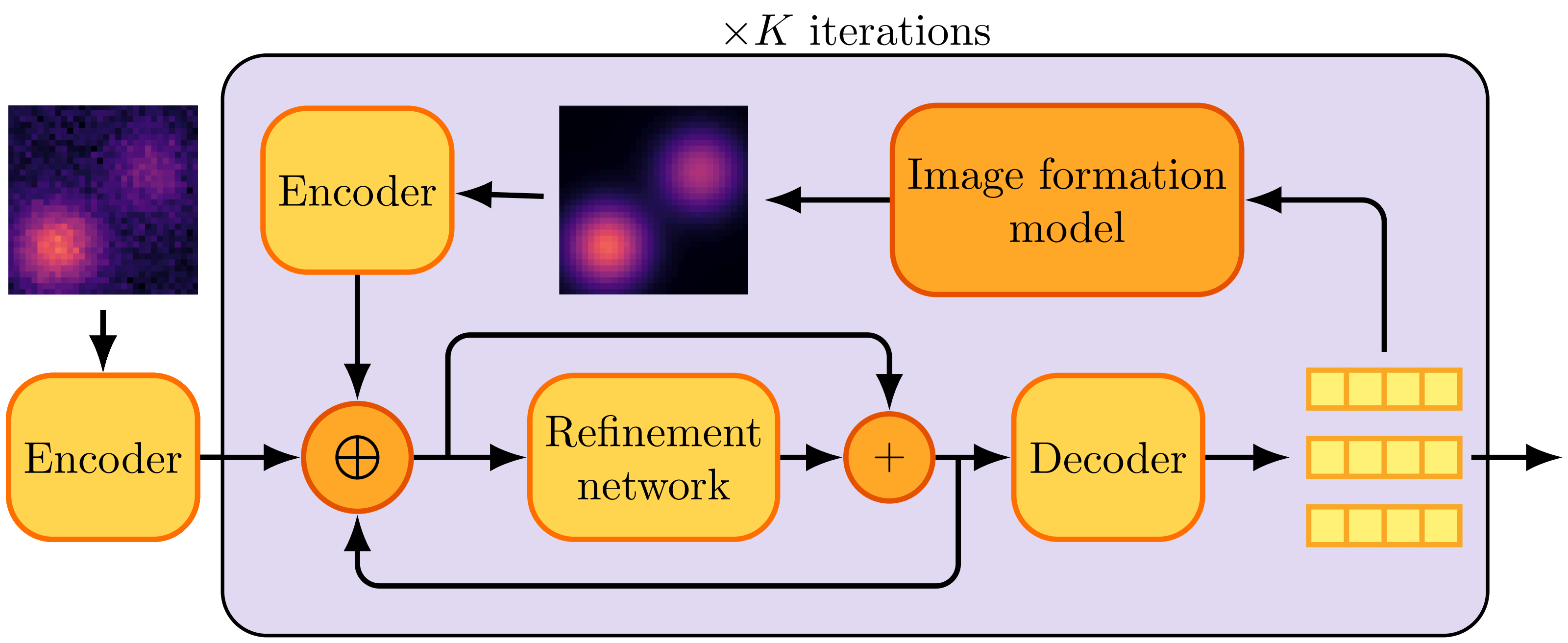
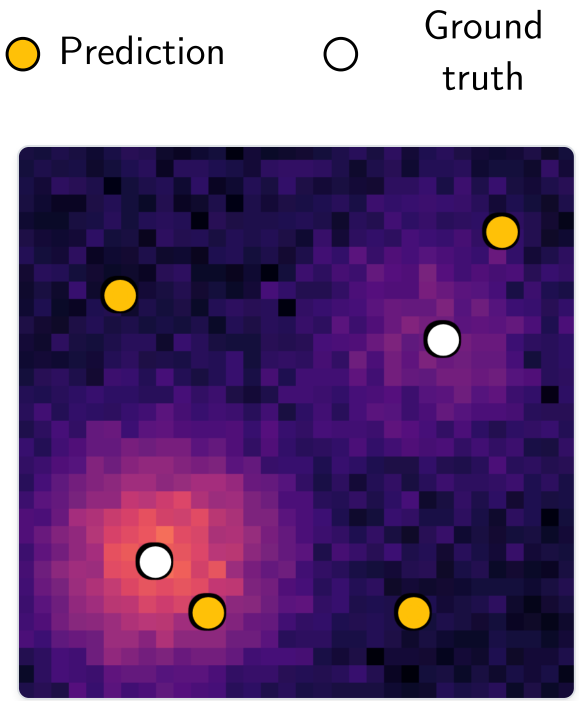


This iterative refinement scheme runs for $K = 3$ iterations in practice.

Loss function

We have constructed a loss function using **regularized-bipartite maching**.



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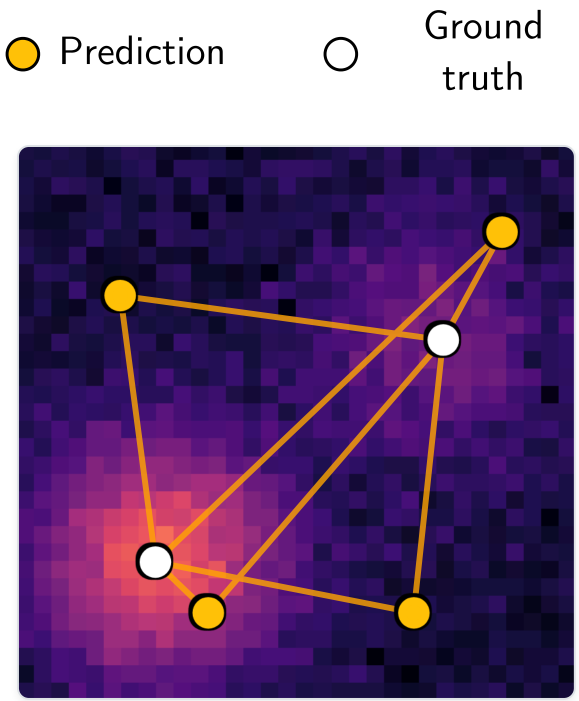

- Compute pairwise localization costs (with negative normal log-likelihood):

$$l_{i,j} = (\hat{\boldsymbol{x}}_i - \boldsymbol{x}_j)^\top \boldsymbol{\Sigma}^{-1} (\hat{\boldsymbol{x}}_i - \boldsymbol{x}_j) + \log\det(\boldsymbol{\Sigma})$$

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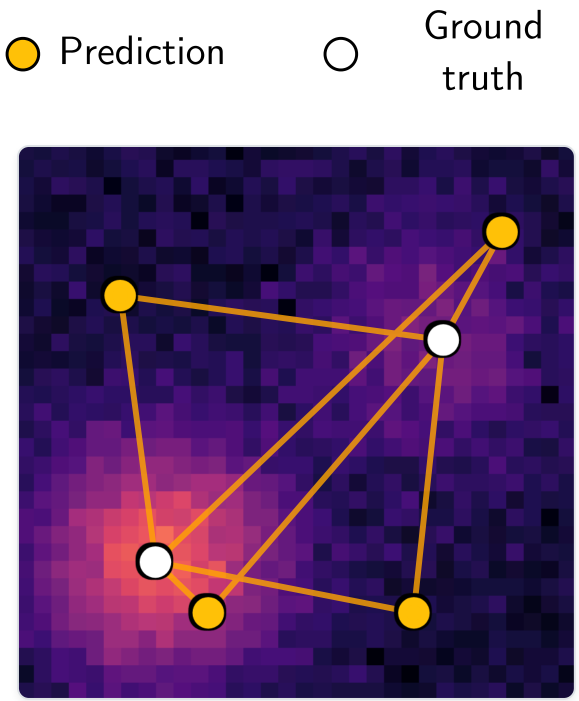

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- Create a squared cost matrix with localization cost and detection penalty (with binary cross-entropy):

$$\mathbf{C} = \begin{bmatrix} l_{1,1} - \log(\hat{s}_1) & l_{2,1} - \log(\hat{s}_2) & l_{3,1} - \log(\hat{s}_3) & l_{4,1} - \log(\hat{s}_4) \\ l_{1,2} - \log(\hat{s}_1) & l_{2,2} - \log(\hat{s}_2) & l_{3,2} - \log(\hat{s}_3) & l_{4,2} - \log(\hat{s}_4) \\ -\log(1 - \hat{s}_1) & -\log(1 - \hat{s}_2) & -\log(1 - \hat{s}_3) & -\log(1 - \hat{s}_4) \\ -\log(1 - \hat{s}_1) & -\log(1 - \hat{s}_2) & -\log(1 - \hat{s}_3) & -\log(1 - \hat{s}_4) \end{bmatrix}$$

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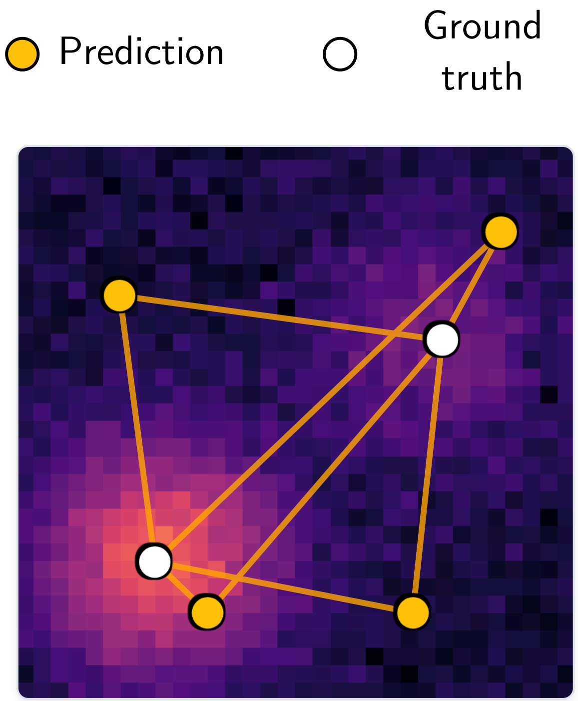

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$$\mathcal{L}(\hat{X}, X) = \langle \Gamma^* \mid C \rangle_{\mathcal{F}}, \quad \text{where } \Gamma^* = \arg\min_{\Gamma \in \mathbb{B}} \langle \Gamma \mid C \rangle_{\mathcal{F}} - \epsilon H(\Gamma),$$

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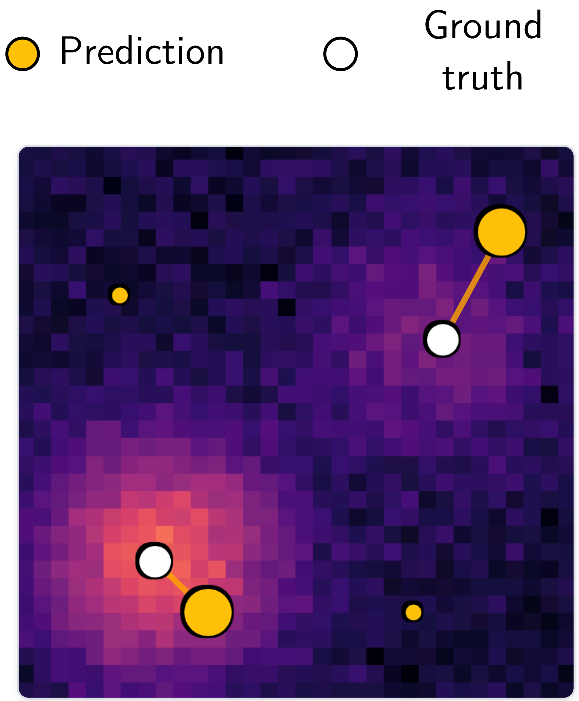

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Synthetic results

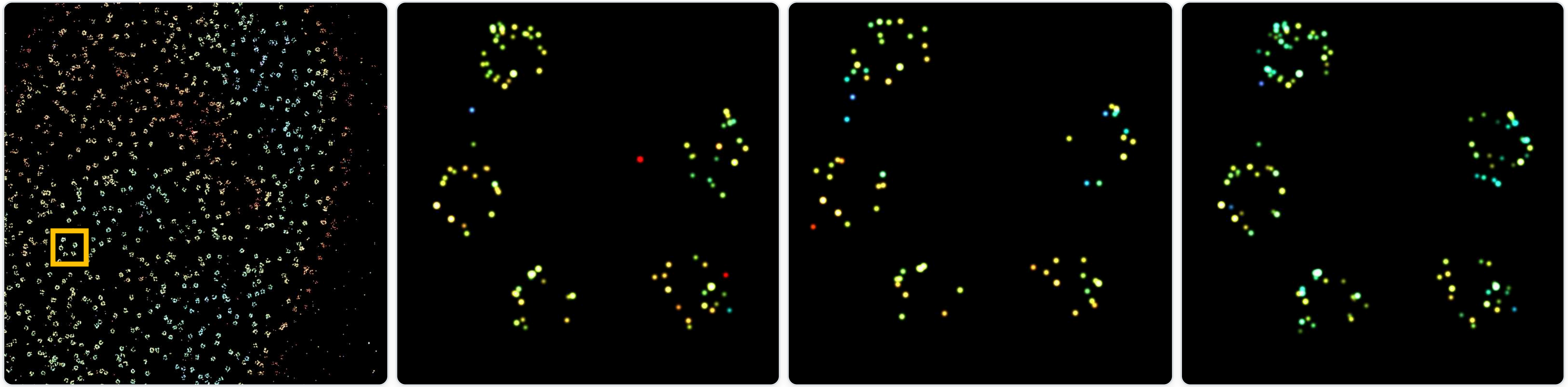
Density	SNR	Method	Precision ↑	Recall ↑	Jaccard ↑	RMSE_{lat} ↓	RMSE_{ax} ↓	E_{3D} ↑
0.2	High	3D-DAOSTORM	0.964	0.919	0.914	11.9	16.9	0.821
		DECODE	0.961 ± 0.003	**0.998 ± 0.001**	0.959 ± 0.003	8.8 ± 0.1	10.7 ± 0.1	0.895 ± 0.003
		LiteLoc	0.996 ± 0.002	0.987 ± 0.001	**0.983 ± 0.001**	9.0 ± 0.1	11.7 ± 0.1	0.912 ± 0.001
		Ours	**0.998 ± 0.002**	0.978 ± 0.016	0.980 ± 0.010	**7.5 ± 0.4**	**10.0 ± 3.5**	**0.920 ± 0.007**
	Low	3D-DAOSTORM	0.978	0.835	0.833	19.3	29.8	0.685
		DECODE	0.918 ± 0.002	**0.978 ± 0.001**	0.903 ± 0.002	20.5 ± 0.1	26.2 ± 0.1	0.757 ± 0.001
		LiteLoc	**0.995 ± 0.001**	0.939 ± 0.001	0.934 ± 0.001	**17.0 ± 0.1**	25.0 ± 0.4	0.798 ± 0.001
		Ours	0.985 ± 0.001	0.961 ± 0.001	**0.947 ± 0.001**	18.8 ± 0.2	**24.5 ± 0.1**	**0.802 ± 0.002**
2.0	High	3D-DAOSTORM	0.914	0.678	0.643	56.8	76.6	0.373
		DECODE	0.923 ± 0.003	**0.946 ± 0.002**	0.876 ± 0.004	32.2 ± 0.3	33.0 ± 0.4	0.706 ± 0.004
		LiteLoc	**0.993 ± 0.001**	0.863 ± 0.002	0.858 ± 0.001	30.7 ± 0.2	36.0 ± 0.3	0.699 ± 0.001
		Ours	0.992 ± 0.002	0.895 ± 0.011	**0.883 ± 0.007**	**24.8 ± 0.6**	**28.4 ± 0.5**	**0.750 ± 0.004**
	Low	3D-DAOSTORM	0.914	0.496	0.475	74.4	120.0	0.116
		DECODE	0.859 ± 0.034	**0.874 ± 0.006**	0.756 ± 0.027	56.4 ± 0.3	65.3 ± 0.4	0.468 ± 0.008
		LiteLoc	**0.992 ± 0.001**	0.729 ± 0.002	0.725 ± 0.001	**46.2 ± 0.4**	63.8 ± 0.2	0.500 ± 0.002
		Ours	0.973 ± 0.003	0.812 ± 0.007	**0.794 ± 0.005**	48.4 ± 0.6	**59.5 ± 0.4**	**0.536 ± 0.003**
8.0	High	3D-DAOSTORM	0.910	0.392	0.379	83.3	133.9	0.009
		DECODE	0.973 ± 0.001	**0.627 ± 0.002**	**0.617 ± 0.002**	59.58 ± 0.02	71.5 ± 0.5	0.371 ± 0.006
		LiteLoc	0.988 ± 0.001	0.557 ± 0.003	0.553 ± 0.003	60.4 ± 0.2	80.5 ± 0.5	0.319 ± 0.004
		Ours	**0.989 ± 0.001**	0.578 ± 0.008	0.574 ± 0.008	**52.5 ± 0.4**	**64.3 ± 0.7**	**0.384 ± 0.003**
	Low	3D-DAOSTORM	0.908	0.211	0.217	92.6	175.8	−0.216
		DECODE	0.93 ± 0.03	**0.415 ± 0.005**	**0.402 ± 0.009**	80.25 ± 0.04	105.2 ± 1.0	0.090 ± 0.007
		LiteLoc	0.986 ± 0.001	0.339 ± 0.002	0.338 ± 0.002	76.1 ± 0.1	110.1 ± 0.8	0.055 ± 0.002
		Ours	**0.983 ± 0.002**	0.376 ± 0.008	0.374 ± 0.008	**74.3 ± 0.5**	**99.4 ± 1.1**	**0.103 ± 0.002**

Long story short, **it works** on **synthetic benchmarks** at low and high densities.

Real results

We have compared reconstructions from different SMLM methods.

Ours [full]	3D-DAOSTORM [zoom]	LiteLoc [zoom]	Ours [zoom]

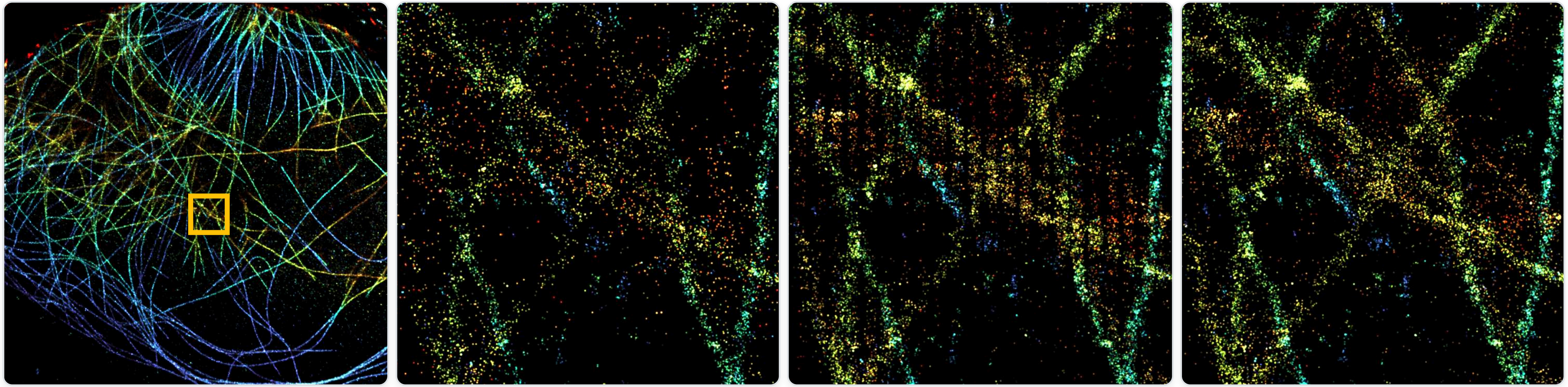


> Our method yields a sharper reconstruction of NPC protein structures.

Real results

We have compared reconstructions from different SMLM methods.

Ours [full]	3D-DAOSTORM [zoom]	LiteLoc [zoom]	Ours [zoom]

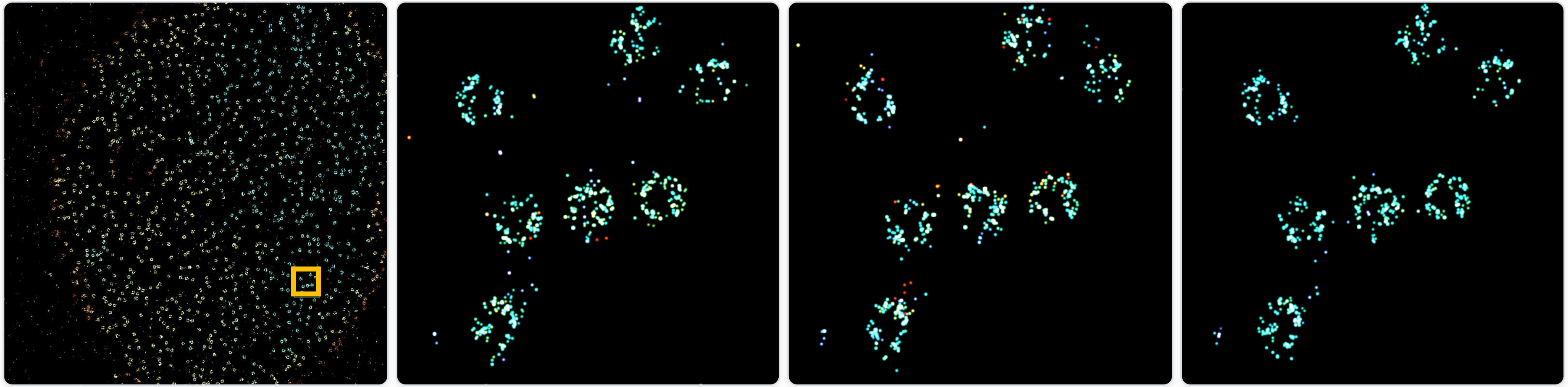


> It also proves efficient with microtubules, and avoids some artifacts present in other non-maximum suppression-based methods.

Real results

We have compared reconstructions from different SMLM methods.

Ours [full] | 3D-DAOSTORM [zoom] | LiteLoc [zoom] | Ours [zoom]



> In these reconstructions, depth is encoded via colors, and our method demonstrates better axial coherency than others.

Limitations

Synthetic **training** and **inference** data **distributions** must be closely **matched**.

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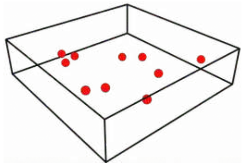

Is a uniform distribution the best option ?

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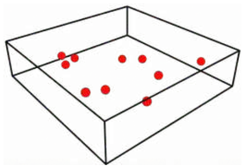

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Temporal dynamics

OFF $\xrightarrow{\text{uniform}}$ ON $\xrightarrow{\tau_{BLEACH}?}$ BLEACH

$\tau_{ON}? \uparrow \quad \downarrow \tau_{DARK}?$

DARK

Are the time constants correct ?

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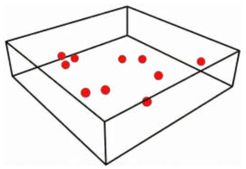

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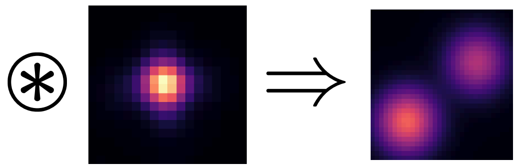
PSF



Is the PSF model accurate ?

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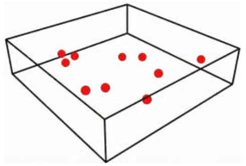

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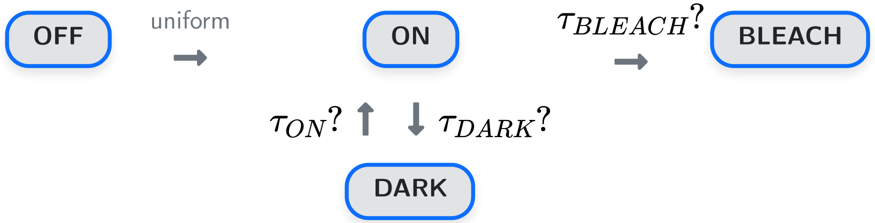

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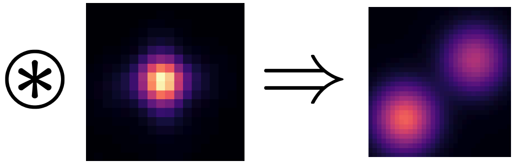
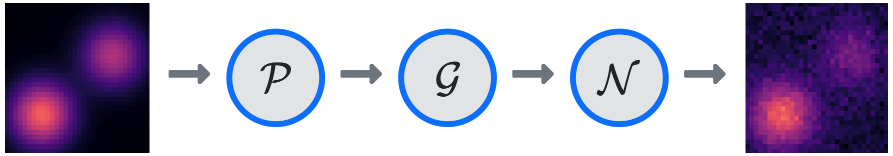


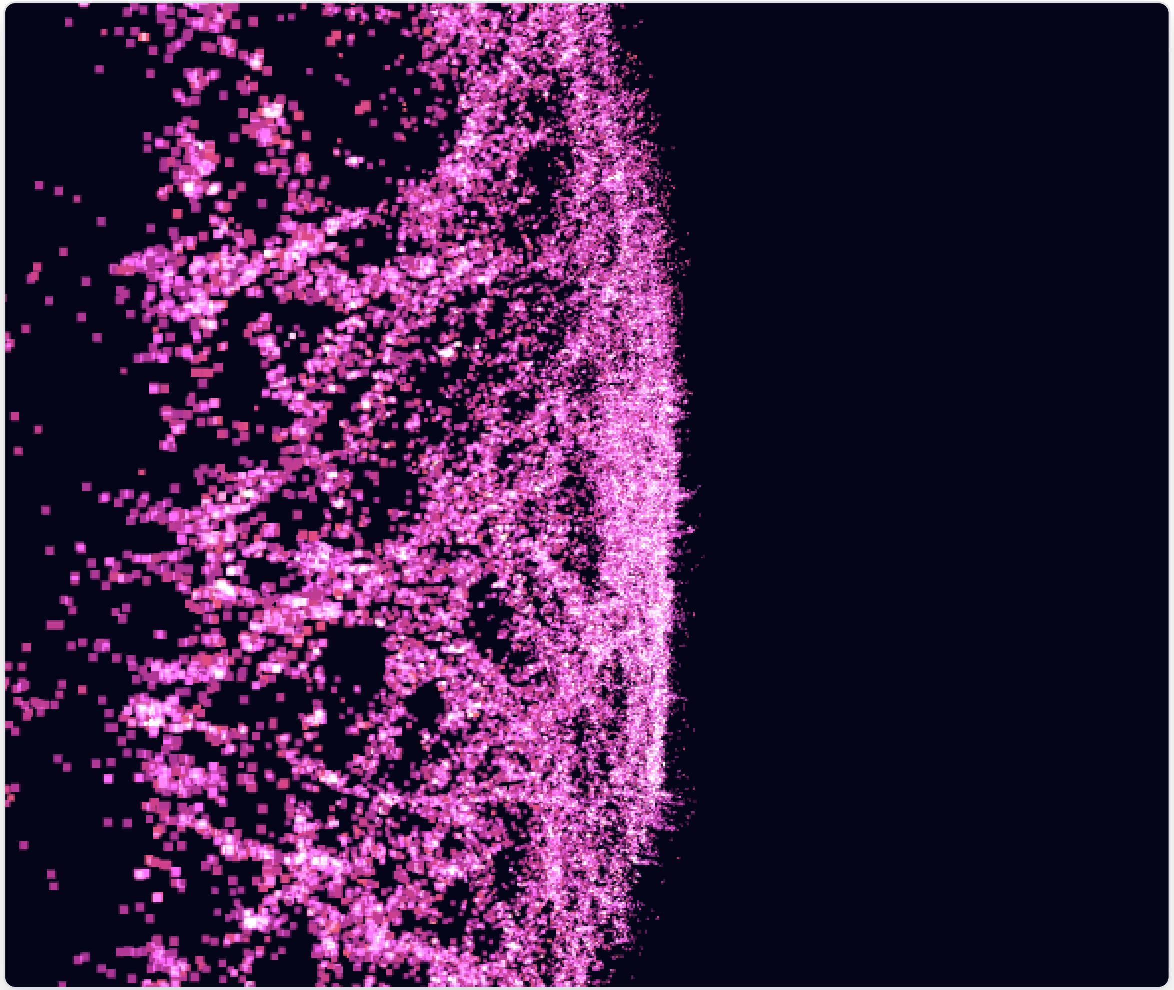
Is the PSF model accurate ?

Camera model



Is the camera model realistic ?

Grazie per l'attenzione



Any questions ?

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